

Product Requirements Document

AI RFP Compliance Auditor - Demo Product

USI Document Intelligence & Verification Layer

Version	1.0
Status	Product specification for demo build
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Demo-only product specification. Use synthetic or public data unless USI provides written authorization for internal documents.

Document Control

Document owner	Riyad Scally
Primary audience	USI executive sponsors, product stakeholders, engineering team, security/privacy reviewers
Product phase	Demo prototype / pre-pilot discovery
Confidentiality	Confidential working document; not a production security assessment
Source inputs	USI AI opportunity assessment; stakeholder discussion with Sales & Marketing and Corporate Strategy & Operations
Decision requested	Approve demo build, review scope, and identify pilot sponsor/data owner

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1. Executive Summary

This PRD defines a demo product that makes AI-assisted RFP review auditable, evidence-based, and safe for human decision-making.

Product thesis. USI does not primarily need another system that writes convincing proposal text. It needs a verification layer that can prove what the source RFP requires, detect missing submission items, and flag unsupported claims before a bid is submitted.

North-star outcome: A reviewer can answer “Are we missing anything, and can every critical output be traced to the source?” in minutes rather than relying on a persuasive but potentially incorrect AI draft.

Working product name	AI RFP Compliance Auditor
Platform name	USI Document Intelligence & Verification Layer
Primary users	Sales/proposal leads, compliance reviewers, operations contributors, executive approvers
Demo objective	Show that an evidence-first AI workflow can identify missing forms, unsupported claims, conflicting requirements, and submission blockers
Primary success moment	The demo catches five intentionally missing forms and traces every flagged requirement to an exact page/section
Recommended demo length	4-6 minutes

2. Product Context and Opportunity

2.1 Business context

- USI responds to document-heavy commercial and public-sector opportunities where submission completeness, insurance terms, forms, deadlines, staffing obligations, and formatting requirements can determine eligibility.
- Stakeholders have already experimented with AI-assisted RFP review and drafting, but experienced unreliable outputs, including invented requirements, cross-document contamination, and missed forms.
- A plausible-looking AI response can increase risk because reviewers may spend more time validating it than they would spend reading the source manually.
- The product opportunity is therefore a reliability and auditability problem before it is a text-generation problem.

2.2 Problem statement

Problem: Proposal teams need speed, but they cannot safely trust AI-generated extraction or drafts when a single missed form, false requirement, or unsupported claim could disqualify a bid or create contractual risk.

2.3 Why current approaches fail

Failure mode	What happens	Business consequence
Generic prompting	A user uploads a long RFP and asks for a summary or draft without a controlled schema.	Important requirements are omitted or merged into narrative text.
Cross-document contamination	The model incorporates similar requirements from memory or another source.	The system invents obligations that do not exist in the active procurement.
No evidence trail	Outputs do not preserve page/section citations.	Reviewers cannot quickly verify the result and must reread the entire source.
One-pass generation	The same pass extracts, interprets, and drafts.	Errors compound and are difficult to isolate.

Failure mode	What happens	Business consequence
Automation bias	Polished language appears authoritative.	Teams over-trust incorrect output or miss subtle gaps.
No submission gate	The tool generates content but does not block missing mandatory items.	A proposal can appear complete while required forms remain absent.

3. Product Strategy

3.1 Vision

Build a reusable document intelligence and verification layer that begins with RFP compliance and can later support contracts, post orders, onboarding documents, credentials, SOPs, and client reporting.

3.2 Product principles

Principle	Implication for the product
Evidence before eloquence	Every critical extraction and audit finding must link to a specific source page/section.
Human authority	AI recommends and flags; authorized users approve, reject, and make final submission decisions.
Separate extraction from verification	The system uses independent passes and deterministic checks rather than trusting a single model response.
Fail visibly	Low confidence, conflicts, parsing errors, and missing evidence must be shown, not hidden.
Narrow demo, extensible architecture	The demo solves one painful workflow while exposing reusable document, retrieval, evidence, and review components.
No production claims from synthetic testing	Demo results communicate feasibility, not production accuracy or legal compliance.

3.3 Objectives

- Reduce time required to construct an initial requirements register and submission checklist.
- Make every extracted obligation traceable to the source document.
- Detect missing forms, deadlines, acknowledgments, attachments, and unresolved mandatory items.
- Audit an AI- or human-generated draft for unsupported claims and conflicts with the RFP.
- Provide a clear submission-readiness view with explicit human approval gates.
- Demonstrate an architecture that could be reused across other USI document-heavy workflows.

3.4 Non-goals

- Autonomously submitting bids or communicating with customers.
- Producing final legal, insurance, pricing, or staffing decisions.
- Replacing proposal managers, compliance reviewers, operations, finance, or counsel.
- Integrating with production USI systems during the demo phase.
- Using confidential USI documents without written authorization and approved handling controls.
- Guaranteeing that the system will find every requirement in every document format.

4. Users and Stakeholders

Role	Primary need	Product responsibility
Proposal / sales lead	Understand requirements quickly and coordinate a compliant response.	Upload documents, review extracted requirements, assign owners, manage

Role	Primary need	Product responsibility
		submission checklist.
Compliance reviewer	Confirm mandatory items and detect hidden or conflicting obligations.	Validate citations, classify risks, approve critical requirements.
Operations contributor	Review staffing, transition, supervision, training, and site obligations.	Confirm operational feasibility and provide source-backed responses.
Finance / insurance reviewer	Validate pricing attachments, insurance thresholds, bonds, and commercial terms.	Approve domain-specific items; mark exceptions.
Executive approver	Know whether the bid is ready and where material risk remains.	Review readiness report and approve final gate.
Demo operator	Present the concept without exposing real data.	Load synthetic/public documents and walk through the prepared scenario.

4.1 Stakeholder map

- Executive sponsor: confirms strategic relevance and authorizes a pilot.
- Business owner: owns the RFP process, data access, reviewer participation, and success metrics.
- Technical owner: approves architecture, vendors, credentials, hosting, and model access.
- Security/privacy reviewer: approves document handling, retention, external processors, and logging.
- End-user reviewers: validate whether outputs are accurate, complete, and faster than the current process.

5. Scope and Release Definition

5.1 Demo release: Must have

Epic	Included capability	Priority
Document ingestion	Upload one primary RFP plus optional addenda and supporting files; parse pages and preserve source metadata.	P0
Requirements register	Extract structured obligations with category, mandatory status, deadline, source quotation, page, section, and confidence.	P0
Evidence viewer	Open the exact source page from any requirement and highlight supporting text.	P0
Submission checklist	Track required forms, signatures, acknowledgments, insurance, attachments, events, and unresolved items.	P0
Draft compliance audit	Upload a draft response and flag unsupported claims, contradictions, missing responses, and uncited assertions.	P0
Readiness report	Show blockers, warnings, human approvals, and an exportable review summary.	P0
Audit log	Record analysis runs, model/config version, user decisions, and export events.	P1
Assignment/status	Assign requirement owner and mark unreviewed, confirmed, exception, or not applicable.	P1

5.2 Should have

- Addendum comparison that shows requirements added, removed, or changed.

- Bulk filter by requirement category, owner, status, confidence, or severity.
- Requirement deduplication and cross-reference detection.
- Export to CSV and a concise PDF/Word readiness report.
- Synthetic “known answer” test mode for demo quality assurance.

5.3 Later / pilot candidates

- Approved USI content library and reusable proposal language.
- Integration with SharePoint, CRM, proposal software, or contract repository.
- Collaborative comments, notifications, and workflow deadlines.
- Role-specific copilots for operations, finance, insurance, and legal reviewers.
- Cross-bid analytics, win/loss learning, and reusable compliance patterns.

6. User Journeys and Experience

6.1 Journey A - Analyze a new RFP

1. User creates a workspace and enters opportunity name, customer/agency, response deadline, and internal owner.
2. User uploads the RFP and any addenda. The product displays upload status, page count, and parsing warnings.
3. The analysis pipeline extracts requirements and builds a structured requirements register.
4. User filters mandatory items and opens a requirement to view the source page and highlighted supporting text.
5. User confirms, edits, rejects, or marks each critical item as needing expert review.
6. Product builds a submission checklist and highlights missing forms, signatures, deadlines, and acknowledgments.
7. User exports the register or proceeds to audit a draft response.

6.2 Journey B - Audit a proposal draft

8. User uploads or pastes a draft proposal response.
9. The system segments the draft into claims and response sections.
10. Each claim is matched against the RFP and approved reference materials.
11. The product flags unsupported claims, contradictions, missing requirement responses, and statements requiring human proof.
12. User reviews each finding, opens the relevant source, and records a resolution.
13. The readiness score updates only after critical blockers are resolved or explicitly accepted by an authorized reviewer.

6.3 Journey C - Executive readiness review

14. Executive opens a one-page summary of completeness, risk, and unresolved approvals.
15. Product displays critical blockers first, followed by warnings and low-confidence items.
16. Executive can drill into evidence but is not required to inspect every extraction.
17. Final status remains “Not ready” until all mandatory gates are approved or an exception is documented.

7. Functional Requirements

ID	Capability	Requirement	Priority
FR-001	Workspace creation	Create a named opportunity with deadline, owner, customer, and optional description.	P0
FR-002	Document upload	Accept PDF and DOCX for the demo; show file name, size, hash, page count, and parsing status.	P0
FR-003	Addenda grouping	Associate addenda with the primary RFP and preserve document order/version.	P1
FR-004	Page-preserving parsing	Store page-level text, tables where available, coordinates or text anchors, and parser confidence.	P0
FR-005	Structured extraction	Extract requirement type, description, mandatory flag, deadline, responsible function, source quote, page, section,	P0

ID	Capability	Requirement	Priority
		and confidence.	
FR-006	Requirement status	Support unreviewed, confirmed, rejected, exception, not applicable, and superseded.	P0
FR-007	Source navigation	From any requirement, open the source page and visually identify the supporting passage.	P0
FR-008	Missing evidence state	Do not present a requirement as verified when no source evidence is available.	P0
FR-009	Checklist generation	Generate checklist categories for forms, signatures, attachments, insurance, bonds, pricing, meetings, deadlines, and acknowledgments.	P0
FR-010	Blocker detection	Mark missing mandatory items, expired deadlines, unacknowledged addenda, and unresolved conflicts as blockers.	P0
FR-011	Draft upload	Accept a proposal draft and parse headings, paragraphs, tables, and claims.	P0
FR-012	Claim-to-source audit	Classify claims as supported, partially supported, unsupported, contradicted, or requires human proof.	P0
FR-013	Cross-document isolation	Constrain evidence to the active workspace documents and identify any claim not grounded there.	P0
FR-014	Human decision capture	Allow reviewers to accept, correct, reject, comment, assign, or waive findings with rationale.	P0
FR-015	Readiness calculation	Calculate readiness from mandatory completeness and human approval rules, not model confidence alone.	P0
FR-016	Export	Export requirements CSV and a PDF/DOCX readiness summary for the demo.	P1
FR-017	Audit history	Record model/config version, timestamps, source hashes, user changes, and approvals.	P1
FR-018	Re-analysis	Allow re-analysis after document changes while preserving prior run history.	P1

8. Business Rules and Reliability Requirements

Rule	Definition
BR-01	A mandatory requirement cannot be marked complete without either a linked artifact/response or a documented exception.
BR-02	No AI-extracted requirement is “confirmed” until a human reviewer approves it.
BR-03	Every verified requirement must have source document ID, page number, and supporting quotation or anchor.

Rule	Definition
BR-04	An unsupported or contradicted draft claim is a warning or blocker based on configured severity; it is never silently rewritten.
BR-05	The system must distinguish "not found" from "not required."
BR-06	Addenda take precedence over superseded language; conflicts must be shown explicitly.
BR-07	Readiness cannot be 100% while critical parsing errors, unresolved mandatory items, or unreviewed blockers exist.
BR-08	Model confidence is advisory; deterministic checks and human approval govern final status.
BR-09	The demo must never transmit real confidential USI data without approved environment and authorization.
BR-10	The product must disclose that outputs are decision support and not legal, insurance, or contractual advice.

8.1 Reliability targets for the demo

Measure	Demo target	How measured
Known mandatory form recall	100% on curated demo document	Seed document with 10 known forms; verify all are detected.
Citation validity	100% for displayed verified findings	Each clickable citation resolves to correct document/page and supports the statement.
Unsupported-claim detection	At least 90% on curated draft	Insert known unsupported and contradictory claims.
False requirement rate	0 critical false requirements in prepared demo	Use expected-answer fixture and human verification.
Analysis completion	Under 3 minutes for prepared document set	Measured from upload completion to results ready.
Demo stability	Three consecutive successful rehearsals	No uncaught errors; reset flow documented.

9. UX and Content Requirements

9.1 Core screens

Screen	Purpose	Required elements
Opportunity list	Select or create an analysis workspace.	Name, customer, deadline, owner, status, last analysis.
Upload and processing	Load documents and understand processing state.	File list, document type, progress, parser warnings, retry.
Overview dashboard	Show the state of the submission.	Readiness, blockers, warnings, high-risk clauses, missing items, review progress.
Requirements register	Review and manage structured obligations.	Filters, category, mandatory status, owner, deadline, confidence, review status.
Evidence viewer	Verify a finding against the source.	Document/page selector, highlighted passage, extracted field, approve/edit/reject.

Screen	Purpose	Required elements
Submission checklist	Track tangible submission items.	Category, item, evidence/artifact, owner, status, due date, blocker flag.
Draft audit	Review claims and response coverage.	Finding severity, claim text, classification, matched source, suggested action.
Readiness report	Support final decision and export.	Executive summary, blockers, exceptions, approvals, run metadata, disclaimer.

9.2 UX principles

- Blockers are visually dominant; model-generated prose is secondary.
- Source evidence is one click away from every critical finding.
- Confidence is displayed with an explanation and never used as a substitute for evidence.
- The interface uses explicit states such as “Needs review,” “Source not found,” and “Conflict detected.”
- Users can correct AI output without rerunning the entire analysis.
- The product avoids language that implies autonomous approval or guaranteed completeness.

10. Demo Scenario and Script

10.1 Prepared fixture

- A synthetic or public security-services RFP of approximately 35-60 pages.
- Two addenda, one of which changes a deadline and adds a required acknowledgment.
- Ten known mandatory submission forms, with five intentionally omitted from the draft package.
- A draft response containing three unsupported company claims, two contradictory staffing statements, and one incorrect insurance threshold.
- A known-answer file used only for internal demo testing.

10.2 Five-minute demo flow

Time	Action	Message to audience
0:00-0:30	Open the opportunity dashboard.	“The goal is not to write a prettier proposal. It is to know whether the proposal is safe to submit.”
0:30-1:15	Show documents and completed analysis.	“The system preserves the primary RFP and addenda as isolated, versioned sources.”
1:15-2:05	Open the requirements register and click a mandatory form.	“Every verified requirement has page-level evidence; nothing is accepted because the model merely said so.”
2:05-2:50	Open the checklist and reveal five missing forms.	“This is the specific failure mode the demo is designed to prevent.”
2:50-3:45	Open draft audit findings.	“The product separates unsupported, contradicted, and human-proof claims instead of silently rewriting them.”
3:45-4:30	Resolve one item and show readiness update.	“Readiness is driven by review rules and approvals, not model confidence.”
4:30-5:00	Export the readiness report and explain extensibility.	“The same verification layer can later support contracts, post orders, onboarding, credentials, and SOPs.”

Do not say: “The AI guarantees compliance” or “This replaces proposal review.” The demo proves an evidence-first workflow, not perfect automated interpretation.

11. Metrics and Acceptance Criteria

11.1 Product success metrics

Metric	Definition	Demo/pilot direction
Time to first requirements register	Elapsed time from completed upload to reviewable structured output.	Demo <3 minutes; pilot benchmark against manual process.
Reviewer verification time	Time to confirm or reject extracted critical items.	Decrease because citations open exact source pages.
Mandatory-item recall	Share of known mandatory items identified.	Demo 100% on fixture; pilot target established with business owner.
False critical findings	Critical requirements shown without supporting source.	Demo zero on fixture.
Blocked submission defects	Missing forms or contradictions found before finalization.	Track count and severity during pilot.
Adoption / trust	Reviewers willing to use output as first-pass decision support.	Qualitative interviews and repeat usage.
Economic value	Hours saved plus avoided bid disqualification/rework.	Estimate after a real shadow-mode pilot.

11.2 Release acceptance criteria

ID	Acceptance criterion
AC-01	Prepared RFP and addenda upload without error and preserve page-level text.
AC-02	The system produces the expected ten mandatory form requirements from the fixture.
AC-03	Each mandatory requirement opens the correct source page and supporting passage.
AC-04	The checklist shows the five intentionally missing forms as critical blockers.
AC-05	The draft audit flags all three unsupported claims and both contradictions in the prepared scenario.
AC-06	A reviewer can correct an extracted field and the audit log records the change.
AC-07	Readiness remains "Not ready" while any critical blocker is unresolved.
AC-08	A final readiness report exports with source references, unresolved items, approvals, run date, and disclaimer.
AC-09	The demo can be reset and repeated without manual database editing.
AC-10	No real USI confidential data is used in the demo environment.

12. Risks, Dependencies, and Governance

Risk	Impact	Mitigation
Complex PDFs / scanned pages	Parser may miss text, forms, or tables.	Show parser confidence; OCR fallback; preserve page images; manual correction.
Hallucinated extraction	False obligations may mislead reviewers.	Constrained schema, source quotation required, independent verifier, human confirmation.
Missed requirements	Incomplete checklist may create false confidence.	Known-answer tests, multiple extraction strategies, low-confidence queue, explicit non-guarantee.
Sensitive bid data	External processing may violate policy or customer terms.	Synthetic/public data for demo; vendor/security approval before internal pilot; encryption and retention controls.
Overbuilding before validation	Time is spent on integrations instead of proving value.	Demo uses isolated upload workflow and curated scenario.
Executive misinterpretation	Prototype is treated as production-ready.	Visible "Demo" label, limitations, readiness checklist, and pilot prerequisites.
No business owner	Project stalls after demo.	Require sponsor, process owner, reviewer cohort, and pilot decision date.
Unclear budget / scope	Interest does not convert to project.	Present fixed pilot options with deliverables, duration, and decision gates after demo.

12.1 Pilot prerequisites

- Named executive sponsor and business process owner.
- Approved data classification and document-handling rules.
- Sample historical RFPs and known submission outcomes, redacted where necessary.
- Access to 2-4 experienced reviewers for expected-answer labeling and evaluation.
- Agreement on shadow-mode operation: the tool advises; existing review remains authoritative.
- Defined pilot budget, time box, success metrics, and go/no-go decision.

13. Roadmap and Pilot Path

Phase	Duration	Deliverables	Decision gate
Demo prototype	10-15 working days	Prepared fixture, PO screens, evidence links, checklist, draft audit, readiness export.	Does leadership see the problem and value clearly?
Discovery + evaluation design	1-2 weeks	Workflow map, data policy, historical sample set, baseline timing, expected-answer rubric.	Is there an approved owner and safe data path?
Shadow-mode pilot	4-6 weeks	Run against selected real RFPs without changing submission authority; measure time and defects.	Does the tool improve speed or quality with acceptable risk?
Controlled production pilot	6-10 weeks	User accounts, integrations, monitoring, support, security review, formal SOP.	Approve scale, redesign, or stop.
Platform expansion	Quarterly	Contracts, post orders, onboarding, credentials, client reports.	Prioritize by ROI and data readiness.

14. Open Questions and Decisions

Question	Owner	Required before
Which team is the formal business owner for the first pilot?	USI leadership	Pilot approval
Which historical RFPs can be used, and what must be redacted?	Business owner + security/privacy	Evaluation dataset
What proposal/RFP tools are already used and should remain authoritative?	Sales/marketing	Integration design
Which document categories are highest risk: forms, insurance, staffing, pricing, legal clauses, or deadlines?	Proposal reviewers	Scoring rules
What model/vendor and data residency requirements apply?	Cyber/security	Architecture approval
Who may approve or waive critical findings?	Process owner	RBAC and readiness rules
What level of accuracy would justify a paid pilot?	Executive sponsor	Go/no-go metrics
Should the initial engagement be internship, project contract, or consulting pilot?	USI leadership/HR	Commercial structure

Final product decision: Approve a demo build only if it remains evidence-first, uses synthetic/public data, and is presented as a decision-support prototype rather than a production compliance system.